

Python 3.13.13 (tags/v3.13.13:01104ce, Apr 7 2026, 19:25:48) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python313/at2.py =====

=== SCENARIO 1: Drone Delivery (GBFS vs A*) ===

GBFS path: ['Start', 'A', 'Hospital'] | actual cost: 15

A* path: ['Start', 'B', 'Hospital'] | actual cost: 9

-> A* avoids the flooded shortcut because it accounts for real travel cost.

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=== SCENARIO 2: Traffic Signal Optimization ===

Initial plan: [50, 17, 11, 57, 27] | cost: 3738.50

Hill Climbing -> plan: [30, 32, 31, 32, 32] | cost: 16.00

Simulated Annealing -> plan: [30, 32, 31, 32, 32] | cost: 16.00

Genetic Algorithm -> plan: [32, 30, 31, 32, 33] | cost: 21.50

-> GA typically finds the lowest-cost (best) plan for large search spaces.

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=== SCENARIO 3: Mars Rover Online Search Agent ===

Final path to goal: [(0, 0), (1, 0), (2, 0), (3, 0), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4), (3, 4), (2, 4), (1, 4), (0, 4), (0, 3), (0, 2), (0, 1), (0, 2), (0, 3), (0, 4), (1, 4), (2, 4), (2, 3), (2, 2), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (3, 5), (4, 5)]

Total moves (incl. backtracking): 30

-> The rover only 'discovers' obstacles as it senses nearby cells,
updating its internal model and re-planning in real time.

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=== SCENARIO 4: Exam Timetabling (CSP: Backtracking + Forward Checking) ===

Assign Math -> ('Morning', 'Hall A') (forward checking OK: True)

Assign Physics -> ('Afternoon', 'Hall A') (forward checking OK: True)

Assign Chemistry -> ('Morning', 'Hall B') (forward checking OK: True)

Assign English -> ('Afternoon', 'Hall B') (forward checking OK: True)

Final Timetable:

Math -> Morning | Hall A

Physics -> Afternoon | Hall A

Chemistry -> Morning | Hall B

English -> Afternoon | Hall B

>>>

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=== SCENARIO 5: Minimax & Alpha-Beta Pruning ===

Minimax result: 3 (matches document: MAX picks 3)

Alpha-Beta result: 3 | leaf nodes evaluated: 6 (out of 4 total)

-> Alpha-Beta reaches the same optimal decision while pruning the 'X' branch (the 9).

Example evaluation function score: 85

Score = 5*Units + 4*Resources + 3*Territory - 2*EnemyStrength